

Original Article

Coumarin glycosides from *Hydrangea paniculata* slow down the progression of diabetic nephropathy by targeting Nrf2 anti-oxidation and smad2/3-mediated profibrosis



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ABSTRACT

Background: Water extract of *Hydrangea paniculata* (HP) stem, rich in coumarin glycosides, has been demonstrated to have renal protective effect in several experimental kidney injury animal models. Currently, it is under pre-clinical development as a class 5 herbal drug against membranous nephropathy. However, whether it also benefits diabetic nephropathy (DN) is not clear.

Purpose: This study was performed to investigate the protective effect of HP on streptozotocin-induced experimental DN, and further understand its molecular mechanisms.

Methods: In the present study, type 1 diabetes rat model was established by the intraperitoneal injection of streptozotocin. HP was orally administered every day for three months. Biochemical analysis and histopathological staining were conducted to evaluate the renal functions. *In vivo* pharmacokinetic study was conducted to analyse the metabolites of HP with high blood drug concentration. *In vitro* assay using these metabolites was performed to analyse their ability to reduce reactive oxygen species (ROS) production induced under high glucose (HG) condition by flow cytometry. Reverse transcription-polymerase chain reaction was conducted to analyse the mRNA level of nuclear factor (erythroid-derived 2)-like 2 (Nrf2) and IL6 and western blot was performed to analyse the phosphorylation status of smad 2/3 in HK2 cells under TGFβ1 stimulation.

Results: The treatment with HP significantly reduced the blood urea nitrogen and serum creatinine content, and urine albumin excretion in diabetic rats, and increased the creatinine clearance rate. Periodic acid-schiff and methenamine staining and immunohistochemistry revealed that HP also ameliorated glomerulosclerosis and tubular vacuolar degeneration, as well as the deposition of fibronectin and collagen IV in the glomeruli. Pharmacokinetic study results revealed that the major coumarin compounds from HP were metabolised into umbelliferone and esculetin. By *in vitro* assay, umbelliferone and esculetin were found to significantly decrease ROS production induced by HG content, as well as increase the mRNA level of Nrf2. HP and its metabolites also can down-regulate fibronectin secretion in HK2 cells stimulated by TGFβ1 and inhibit smad2/3 phosphorylation. **Conclusion:** HP has beneficial effect on DN by increasing Nrf2 expression and inhibiting TGF-smad signal activation. Further, it can be a novel herbal drug against DN.

Introduction

The global population of people with diabetes is expected to almost double by 2030; aging, urbanisation, and nutrition deprivation further

accelerate the incidence of this health problem globally, including in China (Hu and Jia, 2018; Ma, 2018). Diabetes is associated with microvascular and macrovascular complications. Among them, diabetic nephropathy (DN) is a major microvascular complication, and it is a

Abbreviations: ANOVA, one-way analysis of variance; AUC, area under curve; BUN, blood urea nitrogen; CCr, creatinine clearance rate; DN, diabetic nephropathy; ECM, extracellular matrix; EMT, epithelial-mesenchymal transition; GBM, glomerular basement membrane; GFR, glomerular filtration rate; HG, high glucose; HP, *Hydrangea paniculata*; HPLC, high-performance liquid chromatography; IL-6, interleukin 6; MRT, mean residence time; Nrf2, nuclear factor (erythroid-derived 2)-like 2; PAS-M, periodic acid-Schiff and silver methenamine; ROS, reactive oxygen species; SCR, serum creatinine; STZ, streptozotocin; TBS, tris buffer saline; TCM, traditional Chinese medicine; TGFβ1, transforming growth factor-β1

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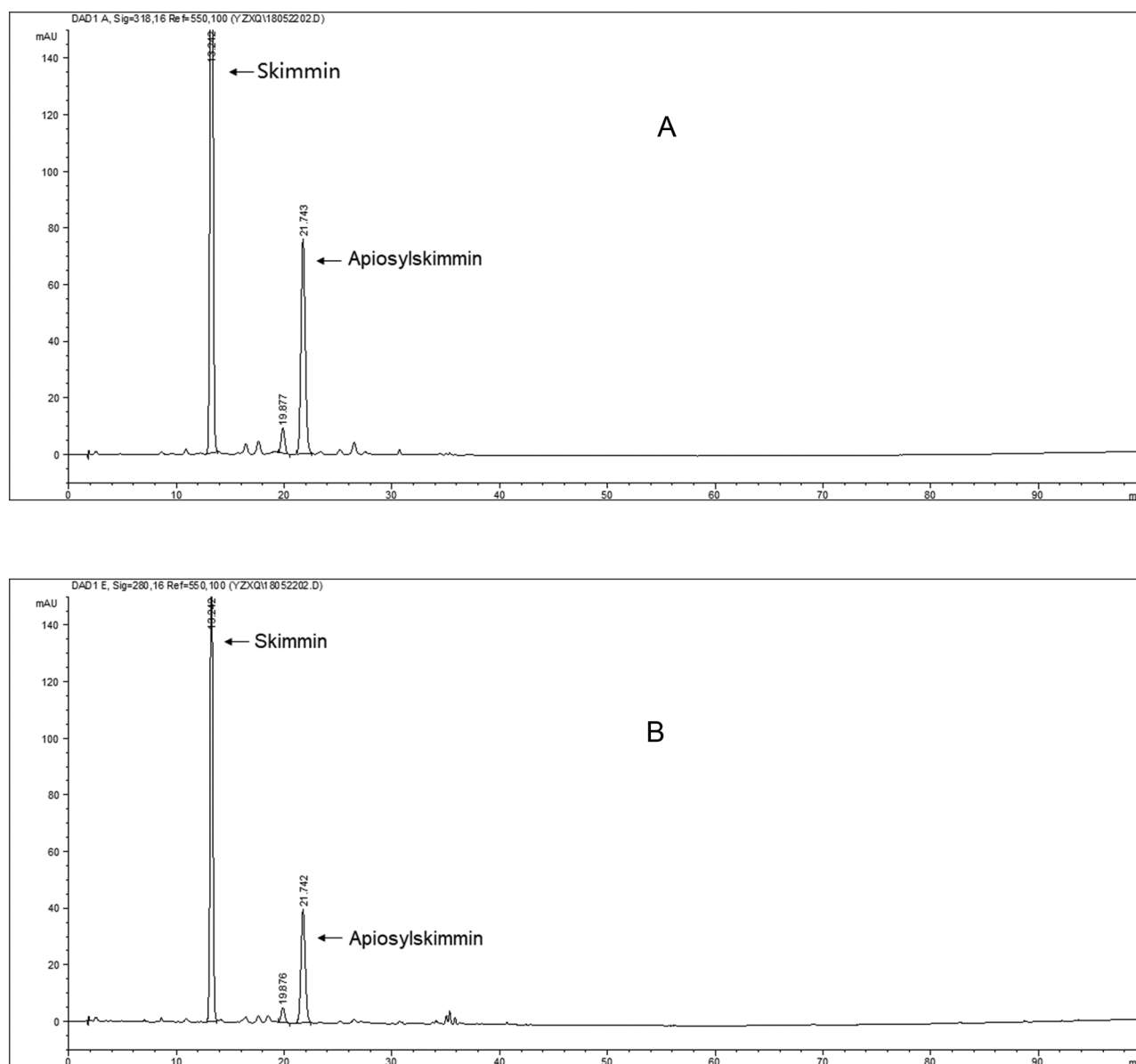


Fig. 1. High-performance liquid chromatography (HPLC)-based chemoprofile of coumarin derivatives from HP. (A) HPLC profile at 318 nm; (B) HPLC profile at 280 nm.

leading cause of end-stage renal disease (Sasaki et al., 2011). It has been reported that approximately 20% of patients with either type 1 or 2 diabetes develop nephropathy after >10 years of diagnosis of diabetes (Soetikno et al., 2014).

Several pathological factors contribute to the progression of DN, including uncontrolled hyperglycaemia, hypertension, intrarenal low-grade inflammation, oxidative stress, formation of advanced glycation end products, and activation of protein kinase C and AMP-activated protein kinases (Han et al., 2017; Soetikno et al., 2014). Among them, transforming growth factor- β 1 (TGF β 1) plays a central role in glomerulosclerosis, which is considered to be the most potent profibrogenic cytokine (Huang et al., 2013). The activation of TGF β 1-smad pathway can stimulate the production of extracellular matrix protein, including fibronectin and collagen IV. These matrix component proteins finally deposit on the glomerular basement membrane (GBM) and increase the thickness of GBM, which is a hallmark of DN (Bottinger and Bitzer, 2002; Meng et al., 2012; Russo et al., 2007). Besides that, under hyperglycaemic condition, glomerular endothelial cells generate excessive reactive oxygen species (ROS), triggering the down-stream renal

damage (Szeto, 2017).

Although DN results in high rate of mortality, there is still lack of effective medicine for clinical treatment. Strict regulation of blood glucose level, combined with anti-hypertension treatment, is still the main-stream treatment strategy. However, even with this treatment, a certain percent of patients eventually develop DN (Zhang et al., 2017b). Therefore, there is need for effective drugs, especially against DN.

For a long time, in China, traditional Chinese medicine (TCM), mainly using herbs as raw material, has been widely used against several chronic diseases. Several herbs have been demonstrated to be effective against diabetes and its complications, such as *Cornus officinalis* (Huang et al., 2018) and *Rhizoma coptidis* (Huang Lian) (Pang et al., 2015). *Hydrangea paniculata* Sieb is one of the folk medicinal herbs, which is widely distributed in the southern part of China. Its stem aqueous extract has been used to treat inflammation, fever, and oedema caused by kidney dysfunction in China for decades. Studies from our lab have focused on *H. paniculata* (HP) for >15 years (Shi et al., 2012, 2010). By in-depth chemical analysis, 90% chemical ingredients of HP extract have been clarified; 75% of which are coumarin

glycosides, among them, skimmin and apiosylskimmin are the major constituents with other ten minor kinds of coumarin glycosides (Sen et al., 2017; Zhang et al., 2017a). Furthermore, it has been reported that water and 20% ethanol extracts of HP stem have high biological activity in LPS-induced acute kidney injury model (Zhang et al., 2017a). The results of our previous study has demonstrated that HP administration has renal protective effect on cisplatin-induced acute kidney injury (AKI) and lipopolysaccharide (LPS)-induced AKI (Sen et al., 2017). Additionally, skimmin administration can slow down the cationised BSA-induced membranous glomerularitis (Zhang et al., 2013). HP is currently under pre-clinical development as class 5 of Chinese medicine against membranous glomerularitis.

Considering the traditional medical usage of HP and its bioactive ingredients, we hypothesised that HP extract might be beneficial on DN, owing to their anti-oxidation and anti-inflammation activities, especially its specific renal protective effect. In the present study, a DN rat model induced by low dose of streptozotocin (STZ) was established to investigate the possible renal protective effect of HP and further elucidate the underlying mechanisms. Furthermore, to understand the major bioactive compounds of HP, which act as anti-DN agent *in vivo*, pharmacokinetic study was also conducted to investigate bioactive metabolites with high blood concentration produced by liver metabolic enzymes and gut bacteria.

Material and methods

HP preparation and chemicals

HP was prepared by the State Key Laboratory of Bioactive Substances and Functions of Natural Medicines, Institute of Materia Medica, Chinese Academy of Medical Sciences, following the extraction procedure described previously (Sen et al., 2017). A high-performance liquid chromatography (HPLC)-based chemoprofile of all coumarin derivatives has been shown in Fig. 1. The major ingredient skimmin was isolated and purified from HP, and its purity was 99.2%. Umbelliferone (catalogue number: H24003) and esculetin (catalogue number: 246573) were purchased from Sigma-Aldrich (St. Louis, MO, USA), and their purities were 99% and 98%, respectively.

Animals and experimental design

Adult male Wistar rats, weighing 180–200 g, were purchased from the SPF Biotechnology Ltd. (Beijing, China). They were housed in a specific-pathogen-free facility with free access to food and water, under constant temperature ($22 \pm 2^\circ\text{C}$) and a daily illumination period of 12 h (7:00 a.m.–7:00 p.m.).

All the rats were randomly allocated to control group ($N = 6$) or diabetic groups. After overnight fast, the rats were subjected to experimental induction of progressive diabetic nephropathy by the injection of STZ (60 mg/kg), which was diluted in citrate buffer solution (0.1 mol/l citric acid and 0.1 mol/l sodium citrate, pH 4.5). Only the rats with blood glucose concentration of >20 mmol/l one week after the induction of diabetes were included in the present study. The rats with diabetes were then randomly administered vehicle buffer (DN control, $N = 9$); 15, 30, and 45 mg/kg HP ($N = 9$); or 10 mg/kg losartan as positive drug ($N = 9$). Both HP and losartan were orally administered every day without any specific interval. Throughout the study, the rats with diabetes were administered 4 U of ultralente insulin (Ultratard HM, Novo Nordisk, A/S, Bagsvaerd, Denmark) every alternate day to maintain their body weight and improve their survival. Every month, the rats were weighed and placed in metabolic cages for the collection of 24-h urine. Blood was collected concurrently from the tail vein to measure the content of blood urea nitrogen (BUN), serum creatinine (Scr), and blood glucose. Both BUN and Scr were measured using commercial assay kit (Nanjing Jiancheng Bioengineer Institute, Nanjing, China), according to the manufacturer's protocol; urinary

albumin concentration was measured using the ELISA kit (Abcam plc, Cambridge, MA, USA). The creatinine clearance rate (CCr) was calculated according to the following formula: $\text{CCr} = \text{Urinary creatinine (mg/ml)} \times \text{urine volume (ml/kg)}/\text{Scr (mg/ml)}$. The whole treatment lasted three months.

At the end of the treatment period, all the rats were anaesthetised with diethyl ether and sacrificed; the renal tissues were sampled for pathological study and protein isolation.

The present study was carried out in accordance with the recommendations of the Chinese Guidelines for Proper Conduct of Animal Experiments, Ethics Committee of Laboratory Animals of Beijing Municipality. The protocol was approved by the Ethics Committee of Laboratory Animals of Chinese Academy of Medical Sciences (approval number 780842).

Histology

The left kidney from all the rats were fixed in 10% phosphate-buffered formalin solution and embedded in paraffin. Sections of 2- μm thickness were cut and stained with Periodic acid-Schiff and silver methenamine (PAS-M) to assess alterations in the glomerular (capillary and mesangium) and proximal convoluted tubules. To characterise glomerular lesions, 10 random glomeruli were checked per rat. Lesions were classified into four severity categories: no lesions, segmental glomerulosclerosis, diffuse sclerosis, and exudative lesion. The proportion (%) of each category was calculated. The vacuolar degeneration of proximal tubular epithelium was characterised semi quantitatively as one of the following four categories: absent, mild, moderate, and severe. Random four $200 \times$ high resolution fields per rat were selected to calculate the severity of tubulointerstitium.

Peripheral blood cell counting

The complete blood cell count was conducted using an automatic haematology analyser (Celltac E MEK-7222; Nihon Kohden, Tokyo, Japan). The number of lymphocytes and neutrophils was counted, and the neutrophil to lymphocyte ratio was calculated for each animal.

Immunohistochemistry

For immunohistochemistry assay of fibronectin and collagen IV, the endogenous peroxidase activity of kidney sections was quenched with 3% H_2O_2 . Antigen retrieval was performed with citrate buffer in a microwave oven. The sections were incubated with primary antibodies (rabbit monoclonal antibody for fibronectin (ab2413) and collagen IV (ab6586), 1:100, Abcam, Cambridge, MA, USA) overnight at 4°C . Labelling was detected with biotin-conjugated goat anti-rabbit IgG (ZSGB-Bio, Beijing, China) and avidin-biotin peroxidase complex ((ZSGB-Bio, Beijing, China). Positive areas were measured using the software Image Pro plus 5.0 (Media Cybernetics, Inc., Rockville, MD, USA) and expressed as percent of positive area per glomerulus. Thirty random cortex $400 \times$ high power fields (HPFs) were selected under ZEISS Observer A1 microscope (ZEISS, Oberkochen, Germany) for positive scoring.

Western blot

To determine the level of TGF β 1 and nephrin in the kidney tissues, renal cortical sections were homogenised using ice-cold RIPA lysis buffer (Applygen Biotechnology, Beijing, China) and quantified by the bicinchoninic acid assay (Applygen Biotechnology) with BSA as standard. Equal amounts of protein were separated by sodium dodecyl sulphate-polyacrylamide gel electrophoresis and electroblotted onto polyvinylidene difluoride membrane. After blocking with 5% (W/V) skim milk solution for 1 h, the membranes were incubated with appropriate antibodies (TGF β 1 (ab92486) and nephrin (ab136894),

1:1000; Abcam) overnight. The membranes were then incubated with horseradish peroxidase-conjugated secondary antibody immunoglobulin G (IgG) at 1:2000 dilution for 1 h at 24–28 °C after being washed thrice using Tris buffer saline with 0.1% Tween 20 (TBS/T). The immunoreactive bands were visualised using an enhanced chemiluminescence system (LAS 4000; GE healthcare, CA, USA). The intensity of the detected bands was analysed using the Image J program.

The total proteins from *in vitro* cultured cells were extracted by the same procedure mentioned above. Smad2/3 (#5678, Cell Signaling Technology Inc., Beverly, MA, USA), p-smad2 (p-Ser465/467, Cell Signaling Technology Inc.), p-smad3 (p-S423 + S425, ab52903, Abcam), and fibronectin (Abcam) were used as primary antibodies in western blot. β -Actin (Santa Cruz Biotechnology, San Diego, CA, USA) was used as the internal standard.

Pharmacokinetic study

The male rats, weighing 180–200 g, were used in the pharmacokinetic study. Briefly, all the rats were fasted for 12 h before administering HP and were fed 4 h after administration. After the oral administration of HP at a dose of 50 mg/kg, approximately 0.2 ml of blood samples was collected in heparinised 1.5-ml polythene tubes by orbital bleeding via capillary tubes at 0.08, 0.25, 0.5, 1, 2, 4, 6, 8, 12, 24, 36, and 48 h post administration. The content of major compounds, skimmin and apiosylskimmin, and their possible metabolites (umbelliferone and esculetin) was determined in rats using a liquid chromatography-tandem-mass spectrometry system, according to the method previously described (Zhang et al., 2017a).

Assay of intracellular ROS induced by high glucose

Rat mesangial cell line HBZY-1 was obtained from the National Infrastructure of Cell Line Resource (Beijing, China) and maintained in low-glucose Dulbecco's modified Eagle's medium (DMEM) (Gibco, Paisley, UK) supplemented with 10% foetal bovine serum (FBS, Gibco, Paisley, UK) and 0.5% normocin (InvivoGen, San Diego, CA, USA). The cells were grown in plastic cell culture dishes with 5% CO₂ at 37 °C and were used in the experiment at 80–90% confluence.

After pre-incubation in DMEM supplemented with 0.5% FBS for 24 h, the cells were treated with low glucose (LG, 1 g/l glucose) and high glucose (HG, 6 g/l glucose) content. In the HG group, HP (50 μ g/ml), skimmin (10 μ M), umbelliferone (10 μ M), and esculetin (10 μ M) were added concurrently, and the same volume of DMSO was also added as vehicle control. After culturing the cells with HG content and 0.5% FBS DMEM for 48 h, the intracellular production of ROS in each group was detected by loading the cultured cells with the fluoroprobe carboxymethyl-H2-dichlorofluorescein diacetate (1 μ M) (CM-H2-DCF-DA, Sigma Chemical Co., St. Louis, MO, USA), a nonpolar compound that is converted into a nonfluorescent polar derivative 2',7'-dichlorofluorescein (DCFH) by cellular esterase after incorporation into cells. Fluorescence intensity was determined immediately with a flow cytometer (excitation λ = 488 nm, emission λ = 515 nm, BD FACSCalibur system, Franklin Lakes, NJ). The average fluorescence intensity was normalised by the total cell protein content of each experimental group of cells as previously described (Yan et al., 2013).

Measurement of Nrf2 and IL6 gene expression by real-time PCR

Rat mesangial cell line HBZY-1 was cultured and treated with HP (50 μ g/ml and 5 μ g/ml), skimmin (10 μ M and 1 μ M), umbelliferone (10 μ M and 1 μ M), and esculetin (10 μ M and 1 μ M) under HG condition in 0.5% FBS DMEM medium, assimilate to ROS assay mentioned above. The LG control and DMSO vehicle HG control were also prepared. After 48 h co-incubation, the total RNA was extracted from cells by Trizol (Life Technology Inc., USA), according to the manufacturer's instructions. The RNA (10 μ g) from each group was then reversed-transcribed

to obtain the cDNA by reverse transcript kit (ReverTra Ace qPCR RT Kit, Toyobo Inc, Tokyo, Japan). The quantitative reverse transcription-polymerase chain reaction (qRT-PCR) was performed using the THUNDERBIRD qPCR Mix (Toyobo Inc, Japan) with the ABI Prism 7900 sequence detector (Applied Biosystems, CA, USA); Δ Ct was adjusted by the house-keeping gene *GAPDH*. The fold change in expression = $2^{\Delta\Delta C_t}$ (Treated - Untreated). The primer sequences of Nrf2 were: forward 5'-GAG ACGCCATGACTGAT-3', reverse 5'-GTGAGGGGATCGATGAGTAA-3'; the primer sequences of IL6 were: forward 5'-ACTTCACAGAGGATAC CAC-3' and reverse 5'-GCATCATCGCTGTTCATAC3'; the primer sequences of GAPDH were: forward 5'- TCCTGCACCACCACTGCTTAG-3' and reverse 5'- AGTGGCAGTGTGGCATGGACT-3'. Real-time PCR cycle conditions were: one cycle of 95 °C for 30 s, 40 cycles of 95 °C for 15 s, 60 °C for 30 s, and 72 °C for 30 s.

Determination of TGF β 1-induced Smad2/3 phosphorylation and secretion of fibronectin

Immortalised human proximal tubular cell line, HK2, was purchased from the American Type Culture Collection (Manassas, VA, USA) and cultured in RPMI-1640 medium (Gibco, Paisley, UK) with 10% FBS and 0.5% normocin. The cells were grown in plastic cell culture dishes with 5% CO₂ at 37 °C. Human recombinant TGF- β 1 (R&D systems, Minneapolis, MN, USA) was dissolved in deionised water supplemented with 0.1% bovine serum albumin and 4 μ M hydrochloric acid. The cells were pretreated with HP (50 and 10 μ g/ml), skimmin (10 and 2 μ M), umbelliferone (10 and 2 μ M), and esculetin (10 and 2 μ M) for 1 h, followed by incubation with TGF- β 1 (10 ng/ml) for 2 h. The cells with TGF- β 1 were then collected for Western blot analysis of phosphorylation of smad2 and smad3.

ELISA to test fibronectin secretion by HK2 under TGF β 1 stimulation

The human Fibronectin ELISA Kit (catalogue number: ab108847, Abcam) was used for quantification of fibronectin in HK2 cell culture supernatants under TGF stimulation. Following the same procedure mentioned above, HK2 cells were pretreated with HP (50 and 10 μ g/ml), skimmin (10 and 2 μ M), umbelliferone (10 and 2 μ M), and esculetin (10 and 2 μ M) for 1 h, followed by incubation with TGF- β 1 (10 ng/ml) for 48 h. After 48 h incubation, the cell culture supernatant was collected for fibronectin analysis, following the instruction in the manual.

Statistical analyses

All data are presented as mean $\pm$ standard error of the mean (SEM), except that pharmacokinetics data, which are expressed as mean $\pm$ standard deviation (SD). The statistical analyses were performed using SPSS version 19.0 for Windows (IBM, Chicago, IL, USA). A comparison of same parameters among all groups was performed by the one-way analysis of variance (ANOVA), with *post-hoc* Tukey's test. For pathological analysis, differences in the severity of glomerulosclerosis and vacuolar degeneration among groups were analysed by Pearson's chi-squared test. The results with *p* values <0.05 were considered statistically significant.

Results

Urine and blood chemistry

The rats with diabetes exhibited reduced body weight gain and elevated plasma glucose level compared with those of the normal control at the beginning of HP treatment. Throughout the treatment period, HP did not show significant glycaemia-lowering effect (Tables 1 and 2).

Although HP did not decrease glycaemia, it significantly improved

Table 1

Blood glucose, BUN, Scr values, etc, after treatment with the HP in STZ-induced Wistar rats for five weeks.

Groups Index	Normal control	Diabetes control	Losartan (10 mg/kg)	HP (15 mg/kg)	HP (30 mg/kg)	HP (45 mg/kg)
N	10	10	10	10	10	10
Body weight (g)	411.2 ± 9.4	239.1 ± 14.8 ^a	268.7 ± 13.8	287.1 ± 14.3 ^c	244.7 ± 19.5	270.3 ± 20.4
Blood Glucose (mM)	5.82 ± 0.13	24.6 ± 1.24 ^b	27.1 ± 1.93	24.6 ± 3.46	25.9 ± 1.52	21.6 ± 3.38
Blood urea nitrogen (mg/l)	21.31 ± 0.83	30.54 ± 1.89 ^b	29.04 ± 1.84	26.31 ± 2.26	23.84 ± 1.64 ^c	21.13 ± 1.60 ^d
Urine albumin (mg/24 h)	0.30 ± 0.10	1.2 ± 0.15 ^b	0.55 ± 0.18 ^d	0.48 ± 0.15 ^d	0.67 ± 0.21 ^c	0.69 ± 0.20 ^c
CCr (ml/min)	0.70 ± 0.16	3.85 ± 0.57 ^b	2.21 ± 0.58 ^d	0.87 ± 0.26 ^d	1.99 ± 0.42 ^c	1.68 ± 0.47 ^d

^a $p < 0.05$.^b $p < 0.01$, compared with the normal control rats.^c $p < 0.05$.^d $p < 0.01$, compared with the diabetes control rats.

renal function deterioration by reducing BUN and Scr content and proteinuria. As shown in Table 1, during the early stage of DN (five weeks post treatment), the BUN and 24 h urine albumin content in diabetic control model increased significantly ($p < 0.01$); the BUN content increased by almost 50% and the urine albumin content increased by five times, compared with those of the normal control. Furthermore, HP administration significantly decreased the BUN and 24 h urine albumin content compared with those in the diabetic control model at different dosages. The concentration of Scr did not change significantly during the early stage of DN after STZ induction for six weeks. Additionally, higher creatinine clearance rate (CCr) was observed in the rats with diabetes compared with that of the normal control during the early DN stage.

After HP treatment for three months, the kidney function in diabetic control group significantly deteriorated. As shown in Table 2, the Scr content increased by 144% and CCr decreased by 85.7% compared with those of the normal control ($p < 0.01$). HP treatment significantly improved the renal function index at different dosages. Losartan administration also presented similar results.

HP attenuates pathological injury in DN rats

The major pathological injuries observed in the rats with DN included glomerular sclerosis, glomerular hypertrophy, segmental and diffuse mesangial expansion of the glomeruli, and vacuolar degeneration in interstitium; all these features were observed in diabetic control rats (Fig. 1). When compared with the diabetic control rats, the HP-treated rats showed significantly less severe sclerosis in glomerulus segments with less percent of segment and diffuse sclerosis and exudative lesions, especially at doses of 30 and 45 mg/kg ($p < 0.01$). However, HP did not exhibit significant effect on diffuse sclerosis (Fig. 2A, Table 3).

Representative photomicrographs of tubular changes are presented in Fig. 2B. Severe vacuolar degeneration in the proximal tubules was found in diabetic control rats. Meanwhile, HP significantly reduced the incidence of tubule vacuolar degeneration and severity level, and showed preventive effect on renal tubule lesions when used at doses of 30 and 45 mg/kg ($p < 0.05$, Fig. 2B and Table 4). Furthermore, HP at doses of 30 and 45 mg/kg showed better beneficial effect on kidney

injuries than losartan at 10 mg/kg.

Expression of fibronectin and collagen IV

The expression of fibronectin and collagen IV in the glomerulus was assessed by immunohistochemistry. Immunostaining of fibronectin and collagen IV was predominantly in the glomeruli area (Fig. 3). In comparison with that of the normal controls, immunostaining for fibronectin and collagen IV was more intense in the kidneys of diabetic rats ($54.2 \pm 7.1\%$ vs. $2.7 \pm 0.6\%$ for fibronectin, $41.7 \pm 5.5\%$ vs. $2.9 \pm 0.7\%$ for collagen IV, and DN control vs. normal control). The treatment with HP significantly reduced the deposition of fibronectin and collagen IV in the rats with diabetes compared with that in the vehicle-treated DN control (Fig. 3, $p < 0.01$ at three dosages of HP). The treatment with losartan also reduced the expression of fibronectin and collagen IV; its effect at a dose of 10 mg/kg was similar to that of HP at doses of 30 and 45 mg/kg.

HP decreased peripheral neutrophil to lymphocyte ratio

As shown in Fig. 4, significantly higher neutrophil to lymphocyte ratio was observed in the DN group than in the normal control (>2 times higher), whereas, HP administration significantly decreased the neutrophil to lymphocyte ratio in a dose-dependent manner. Losartan also decreased the ratio, and its efficiency was similar to that of HP at doses 30 and 45 mg/kg.

HP decreased TGFβ1 protein level and increased nephrin expression in vivo

Western blot of kidney tissue homogenate revealed that the HP treatment significantly reduced the TGFβ1 level in kidney tissues (Fig. 5), which is the most important pro-fibrosis enhancer. On the contrary, the HP treatment significantly increased nephrin expression, which is a biomarker of podocyte, a key cell barrier to prevent albuminuria.

Table 2

Blood glucose, BUN, Scr values, etc, after treatment with the HP in STZ-induced Wistar rats for three months.

Groups index	Normal control	Diabetes control	Losartan (10 mg/kg)	HP (15 mg/kg)	HP (30 mg/kg)	HP (45 mg/kg)
N	10	10	10	10	10	10
Blood glucose (mM)	5.46 ± 0.22	24.05 ± 1.97 ^a	22.6 ± 3.51	18.71 ± 3.04 ^c	18.25 ± 2.27	21.80 ± 4.91
Blood urea nitrogen (mg/l)	7.46 ± 0.82	13.67 ± 1.55 ^a	12.49 ± 0.94	9.81 ± 0.92 ^b	10.37 ± 1.09 ^b	8.83 ± 1.14 ^b
Scr (mg/l)	1.36 ± 0.16	3.33 ± 0.39 ^a	2.25 ± 0.57	2.47 ± 0.42	2.17 ± 0.35 ^b	2.09 ± 0.45 ^b
CCr (ml/min)	3.45 ± 0.62	0.49 ± 0.14 ^a	1.83 ± 0.49 ^b	1.40 ± 0.19	1.77 ± 0.30 ^b	1.53 ± 0.27 ^b

^a $p < 0.01$, compared with the normal control rats; ^b $p < 0.05$, ^c $p < 0.01$, compared with the diabetes control rats.

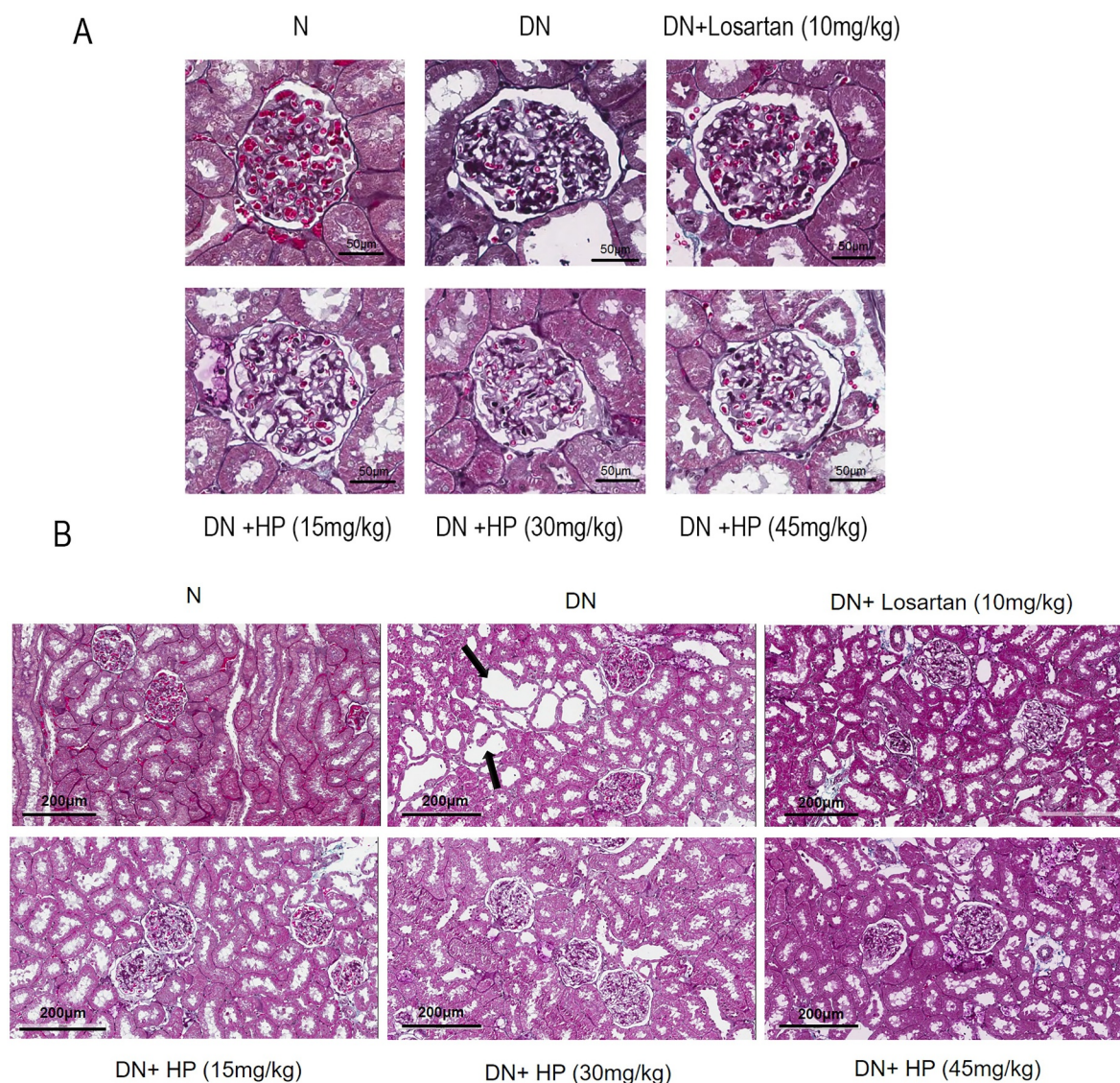


Fig. 2. Histology of kidneys was characterised by staining with PAS. (A) Glomeruli (200 × magnification); (B) Interstitium (400 × magnification).

Pharmacokinetic study of skimmin, apiosylskimmin, and their predicted metabolites

The oral pharmacokinetic profile of plasma skimmin is shown in Fig. 6 and Tables 5 and 6 after a single oral administration at a dosage of 50 mg/kg in Wistar rats. The oral absorbance of skimmin was quick, and it reached the maximum concentration in 0.25–0.5 h. The average plasma drug concentration was 11.64 μM, mean residence time (MRT)_(0–t) was 0.89 h, and area under curve (AUC)_(0–t) was 10.96 μM/h. Two

predicted active metabolites were also detected in the plasma with higher blood concentration. The concentration of umbelliferone reached the maximum after 0.25–0.5 h. The average blood drug concentration was 8.12 μM and AUC_(0–t) was 11.53 μM/h. The concentration of esculetin reached the maximum in 0.5 h; the average blood drug concentration was 2.58 μM and AUC_(0–t) was 3.52 μM/h (Fig. 6, Tables 5 and 6). The concentration of plasma apiosylskimmin after oral administration was too low to be determined.

Table 3

HP effect on glomerulus lesions in diabetic nephropathy in Wistar rats.

Groups	N	Proportion of glomerulus lesions (%)			
		No lesions	Segmented sclerosis	Diffuse sclerosis	Exudative lesion
Normal control	5	96.0 ± 2.45	4.0 ± 5.48	0 ± 0	0 ± 0
DN	6	50.0 ± 1.58 ^b	35.0 ± 6.52 ^b	6.67 ± 8.16	7.33 ± 2.0
Losartan (10 mg/kg)	8	59.5 ± 11.3	29.1 ± 7.49	8.21 ± 5.13	3.21 ± 1.79
HP (15 mg/kg)	7	71.7 ± 4.60 ^c	20.0 ± 8.36 ^d	7.05 ± 3.15	1.25 ± 0.85 ^c
HP (30 mg/kg)	6	67.0 ± 7.48	22.0 ± 6.52 ^c	9.0 ± 4.30	2.0 ± 1.22 ^c
HP (45 mg/kg)	6	70.0 ± 3.53 ^c	20.5 ± 7.58 ^c	7.3 ± 2.55	2.2 ± 2.0

^a*p* < 0.05, ^b*p* < 0.01, compared with the normal control rats; ^c*p* < 0.05, ^d*p* < 0.01, compared with the diabetes control rats, by student's *t*-test.

Table 4
HP effect on renal tubule vacuolar degeneration in Wistar rats.

Vacuolar degeneration (%/(N))							
Groups	N	absent	mild	moderate	severe	χ^2	p
Normal	6	90 (9)	10 (1)	0	0		
DN	9	0 (0)	10 (1)	40 (4)	50 (5)	18.00	<0.001 ^a
Losartan (10 mg/kg)	9	0 (0)	40 (4)	30 (3)	30 (3)	2.433	0.295 ^b
HP (15 mg/kg)	9	0 (0)	40 (4)	50 (5)	10 (1)	4.578	0.101 ^b
HP (30 mg/kg)	9	10 (1)	50 (5)	40 (4)	0 (0)	8.667	0.034 ^b
HP (45 mg/kg)	9	20 (2)	50 (5)	20 (2)	10 (1)	8.000	0.046 ^b

^acompared with the normal control; ^bcompared with diabetes control, *p* value was calculated by Pearson's chi-squared test.

HP and esculentin inhibited ROS production via NRF2 enhancement

As shown in Fig. 7A and B, incubation under HG condition for 48 h significantly increased the intrinsic ROS production in mesangial cells, and HP (50 µg/ml), umbelliferone (10 µM), and esculentin (10 µM) significantly reduced the intrinsic ROS generation (*p* < 0.05 or < 0.01), but skimmion did not alter HG-induced ROS generation.

By qRT-PCR, the mRNA level of Nrf2 and IL6 was examined. Although HP and the three compounds increased the mRNA level of Nrf2 under HG stimulation, we only focused on more than two-fold changes in the present study. As shown in Fig. 7C, esculentin significantly increased the mRNA level of Nrf2 and IL6 dose dependently; both increased the level by more than two folds.

HP and esculentin inhibited TGF-induced fibronectin by inhibiting the phosphorylation of smad2/3

As shown in Fig. 8A, fibronectin was rarely found in normal cell-cultured medium, but its concentration was significantly higher in TGFβ1-induced cell culture medium. HP (50 and 10 µg/ml),

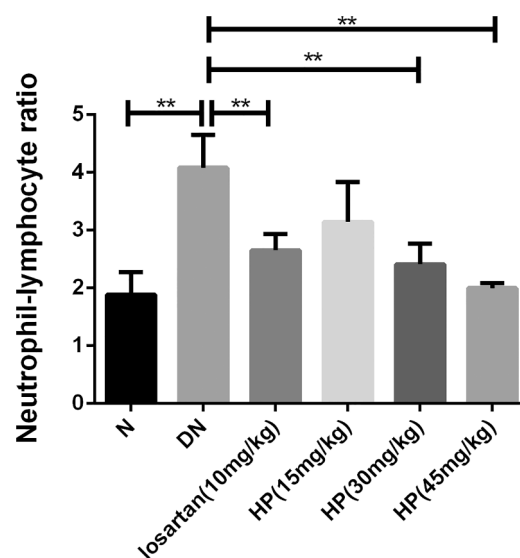


Fig. 4. HP and losartan decrease the peripheral neutrophil-lymphocyte ratio in rats with diabetes. ***p* < 0.01.

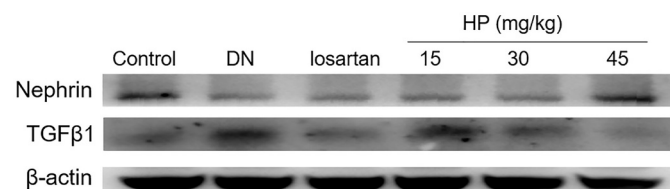


Fig. 5. Western blot of TGFβ1 and nephrin in kidney tissues. HP significantly increased the level of nephrin and decreased the level of TGFβ1 in the kidney tissues of rats with diabetes.

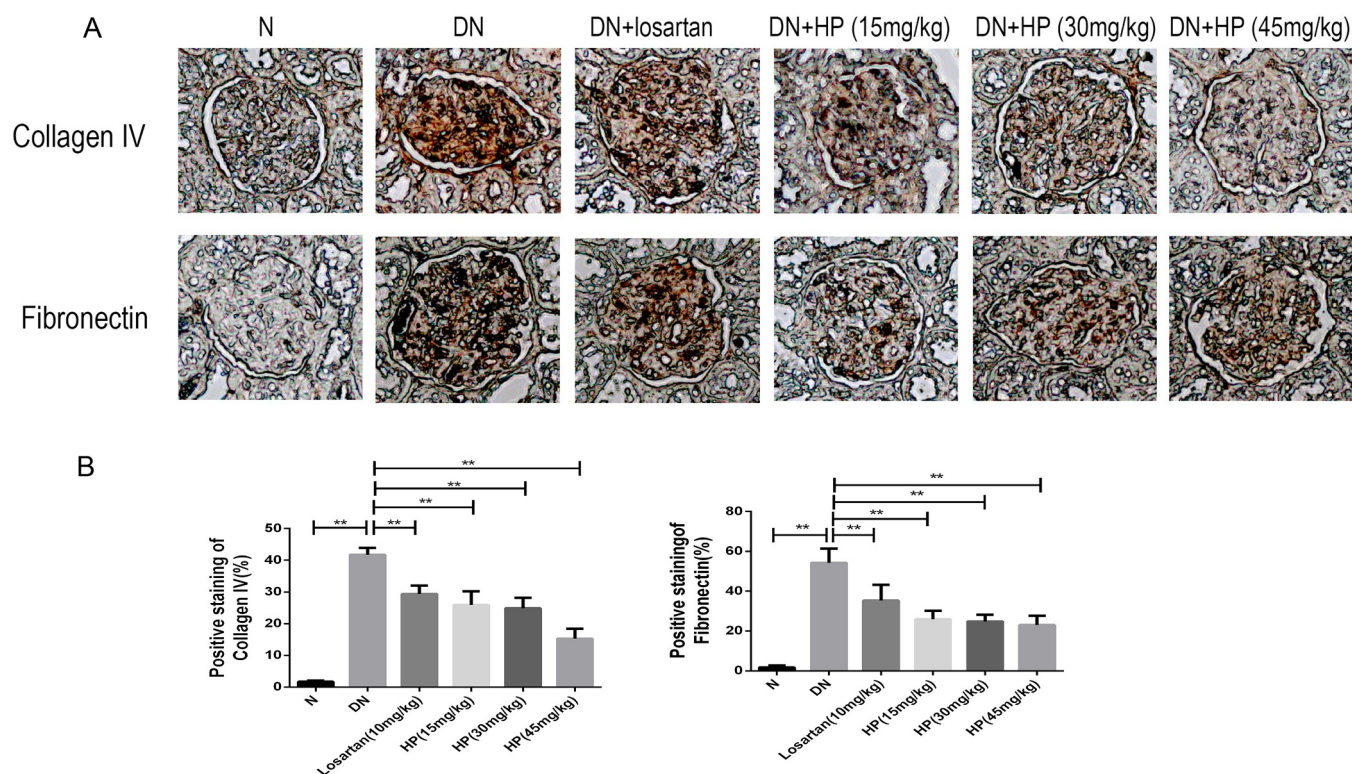


Fig. 3. Immunohistochemistry of fibronectin and collagen IV in the kidney sections. (A) Representative figures of immunohistochemistry (400 × magnification); (B) Quantitation of immunohistochemistry by the Image-pro software. ***p* < 0.01.

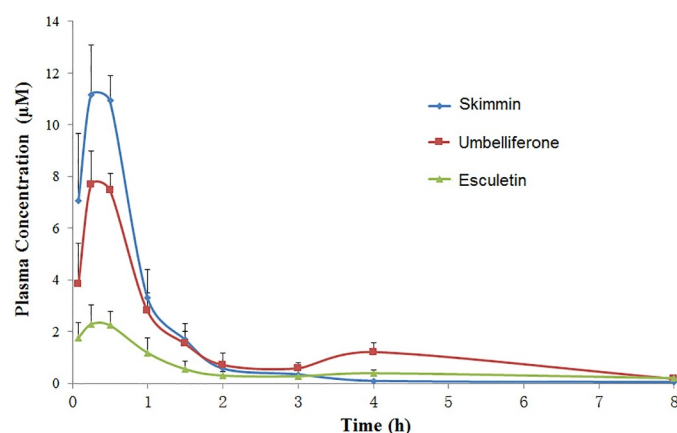


Fig. 6. Plasma concentration-time curve of skimmin, umbelliferone, and esculetin in rats following a single oral dose of 50 mg/kg HP ($n = 5$). The data are expressed as mean $\pm$ SD.

Table 5

Blood drug concentration of major coumarins from HP and their predicted metabolites (skimmin, umbelliferone, esculetin) in rats following a single oral dose of 50 mg/kg at different time points ($N = 5$).

Time (h)	Blood drug concentration (μM)		
	Skimmin	Umbelliferone	Esculetin
0.08	7.07 ± 2.60	3.85 ± 1.58	1.76 ± 0.58
0.25	11.17 ± 1.91	7.68 ± 1.32	2.28 ± 0.75
0.5	10.93 ± 0.98	7.48 ± 0.63	2.24 ± 0.55
1	3.31 ± 1.10	2.80 ± 0.71	1.18 ± 0.59
1.5	1.69 ± 0.62	1.53 ± 0.48	0.55 ± 0.30
2	0.57 ± 0.24	0.71 ± 0.46	0.30 ± 0.16
3	0.34 ± 0.14	0.58 ± 0.21	0.27 ± 0.07
4	0.09 ± 0.02	1.21 ± 0.36	0.39 ± 0.14
8	0.04 ± 0.01	0.17 ± 0.11	0.20 ± 0.06
12	N.A	N.A	N.A

The blood drug concentration after 12-h oral administration is below the detection basic line, so the concentration data is not available (N.A.).

Table 6

Pharmacokinetic parameters of major coumarins and its active metabolites (skimmin, umbelliferone, esculetin) in rats following a single oral dose of 50 mg/kg ($N = 5$).

Index	Unit	Skimmin	Umbelliferone	Esculetin
T_{max}	h	0.40 ± 0.14	0.35 ± 0.14	0.23 ± 0.17
C_{max}	μM	11.64 ± 1.29	8.12 ± 0.54	2.58 ± 0.58
$\text{AUC}_{(0-48)}$	$\mu\text{M}/\text{h}$	10.88 ± 2.06	11.53 ± 1.27	3.52 ± 1.69
$\text{AUC}_{(0-\infty)}$	$\mu\text{M}/\text{h}$	10.96 ± 2.06	12.03 ± 1.14	4.11 ± 2.06
$\text{MRT}_{(0-48)}$	h	0.82 ± 0.05	1.90 ± 0.16	1.44 ± 0.83
$\text{MRT}_{(0-\infty)}$	h	0.89 ± 0.04	2.28 ± 0.37	2.43 ± 1.69
$t_{1/2}$	h	1.43 ± 0.43	1.84 ± 0.47	1.91 ± 1.67

$\text{AUC}_{(0-48 \text{ h})}$, the area under the plasma concentration-time curve from 0 to 48 h; MRT, mean residence time; C_{max} , the maximum plasma concentration; T_{max} , time at maximal concentration; $t_{1/2}$, elimination half-life.

umbelliferone (10 μM), and esculetin (10 and 2 μM) reduced the secretion of fibronectin significantly, especially esculetin. As shown in Fig. 8B, besides down-regulating fibronectin secretion, the level of intrinsic cellular fibronectin and collagen IV was reduced by HP (50 $\mu\text{g}/\text{ml}$), skimmin (10 μM), umbelliferone (10 and 2 μM), and esculetin (10 and 2 μM). The analysis of the downstream effector of TGF β 1 revealed that the expression of both p-smad2 and p-smad3 was significantly reduced by HP, skimmin, umbelliferone, and esculetin with different inhibition capacities; esculetin presented the highest inhibitory effect among all the compounds (Fig. 8B).

Discussion

Renal injury associated with diabetes is the leading cause of end stage renal dysfunction and is becoming more prevalent (Rivero et al., 2009). Proteinuria, enhanced oxidation stress, renal inflammation, and fibrosis contribute to tubulointerstitial injury and the progressive decline in glomerular filtration rate (GFR) (Imig and Ryan, 2013). A previous study from our laboratory suggested that HP extract has anti-oxidation and anti-inflammation bioactivities, therefore, in the present study, we used diabetic animal model to investigate whether HP can slow down the progression of DN as well.

In the present study, after STZ induction of diabetes for five weeks, there was an increase in the CCr of rats with diabetes, indicating the occurrence of glomerular hyperfiltration (Palatini, 2012). Glomerular hyperfiltration generally occurs during the early stages of DN, which is hypothesised to be a precursor of intraglomerular hypertension leading to albuminuria (Amin et al., 2005). Subsequently, the GFR decreases progressively along with the increase in albuminuria, which might, in the long run, lead to end-stage renal failure (Magee et al., 2009; Palatini, 2012). In the present study, we also noticed such pathological progress. During the early five weeks, the GFR of diabetic rats increased significantly compared with that of the normal control, whereas, HP decreased the CCr. After three months of induction of DN, the GFR decreased significantly compared with that of the normal control. By three months of treatment, HP significantly increased the CCr compared with that of the model control, which suggests that HP can improve the renal function. Further, PAS staining also demonstrated that HP can significantly reduce glomerulosclerosis, mesangial expansion, and tubular vacuolar degeneration. The ELISA demonstrated that HP can significantly decrease albuminuria in diabetic rats. Podocyte integrity contributes to proteinuria, and the expression of podocyte integrity biomarker nephrin was significantly increased by HP treatment, which might partially explain that HP can prevent albuminuria (Zhang et al., 2017b).

The peripheral neutrophil to lymphocyte ratio has been considered as a reliable predictive marker of DN in patients with type 2 diabetes in clinical studies (Huang et al., 2015; Khandare et al., 2017). In the present study, this ratio in the model rats with diabetes increased significantly compared with that in the sham control rats, while the intake of HP reversed this increase in a dose-dependent manner. This demonstrates that HP can slow down the progression of DN. On the contrary, several studies have also suggested that this ratio as biomarker is also valuable to predict some other diabetes-related complications, such as arterial stiffness and atherosclerosis in cardiovascular diseases (Balta et al., 2016; Mozos et al., 2017). The results suggest that HP might be valuable for other diabetes-related complications, such as cardiovascular disease, which deserves further in-depth investigation.

To deeply understand the material basis of HP renal protective effect, pharmacokinetic study was performed in the present study. Although apiosylskimmin is the second major compound in HP, it was hardly detected in the plasma of rats. Instead, the two metabolites, umbelliferone and esculetin, were found to in high concentrations in the blood, next only to skimmin. The pharmacokinetic characteristics of rats was similar to the pharmacokinetic parameters of mice (Zhang et al., 2017a); therefore, umbelliferone and esculetin can be potential bioactive compounds responsible for HP pharmacological activity. Through several *in vitro* assays, we demonstrated that esculetin has the highest activity among the three compounds; it can enhance the expression of Nrf2, inhibit TGF β 1-induced phosphorylation of smad2, and further reduce the secretion of fibronectin. Several previous studies have also demonstrated that umbelliferone exhibits antihyperglycaemic effect in STZ-induced diabetic rats (Ramesh and Pugalendi, 2006), and has a beneficial effect on collagen-mediated diabetic polyneuropathy and DN (Ramesh and Pugalendi, 2007). Esculetin also has been reported to attenuate the development of glomerulosclerosis in diabetic rats by interfering with the oxidative stress-related inflammatory

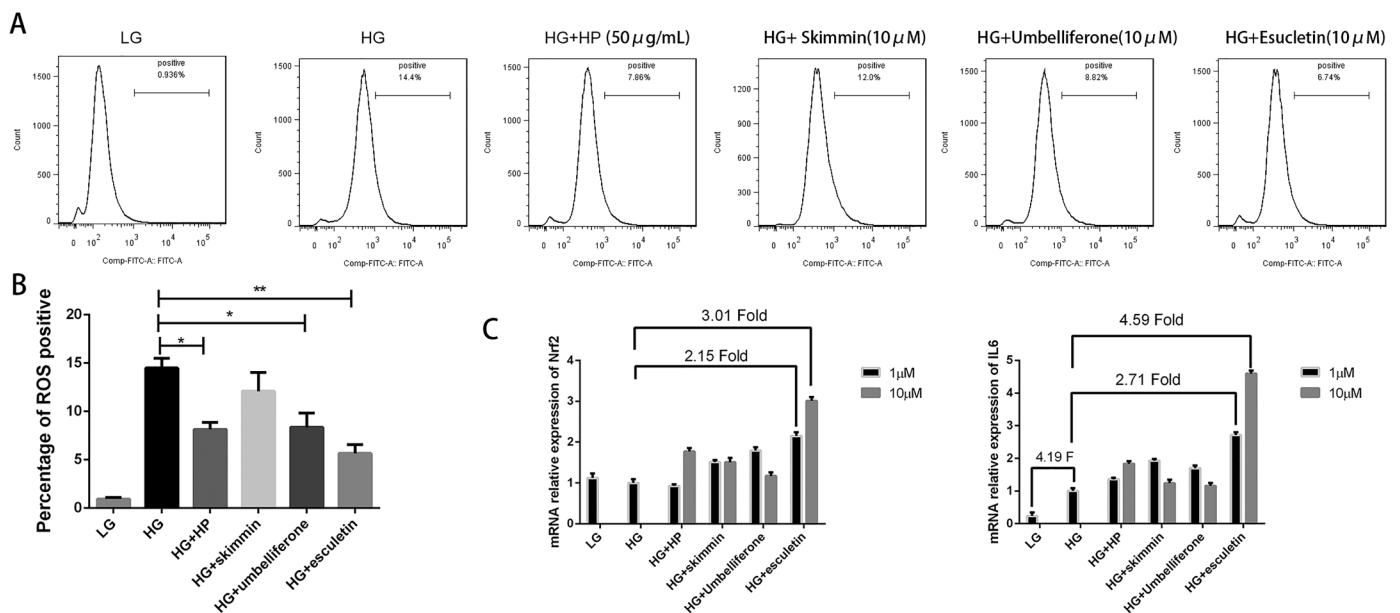


Fig. 7. HP, umbelliferone, and esculletin inhibited high-glucose-induced ROS production in mesangial cells, and qPCR for mRNA level of Nrf2 and IL6. (A) Flow cytometry to test ROS production under HG condition; and (B) quantitation of anti-oxidation effect of HP, skimmin, umbelliferone, and esculletin by flow cytometry; (C) qPCR to analyse the mRNA level of Nrf2 and IL6 under HG condition. The data are expressed as mean $\pm$ SEM of three independent experiments. * p < 0.05, ** p < 0.01.

process (Kadakol et al., 2015), insulin signalling pathway (Kang et al., 2014), and PPAR γ /TGF- β 1 pathway (Surse et al., 2011).

Glomerular mesangial cells are one of main sources of extracellular matrix (ECM) production. Hyperglycaemia-induced oxidative stress is one of major mediators of initiation and progression of DN (Ha et al., 2002). In the present study, HG-induced ROS generation in rat mesangial cells was observed, and umbelliferone and esculletin significantly reduces the ROS content, especially esculletin, suggesting that esculletin has the highest anti-oxidation activity. By real-time PCR, we found that esculletin can significantly increase the mRNA expression of Nrf2 in HG-induced mesangial cells. Nrf2 is considered an important and potential therapeutic target of DN. Nrf2 is a transcription factor that mediates a wide range of adaptive responses to intrinsic and extrinsic cellular stresses (Osburn and Kensler, 2008). Under cellular insult, such as chemical and oxidative stresses, Nrf2 can translocate into the nucleus and bind to the antioxidant response element, in turn inducing the expression of cellular defensive genes, such as those encoding antioxidant enzymes, against oxidative stress. Several studies have reported that the activation of Nrf2 can protect against diabetes-related vascular diseases, including DN (Xue et al., 2008). Using Nrf2 $^{-/-}$ and Nrf2 $^{+/+}$ mice, Jiang et al. demonstrated that Nrf2 conferred protection against hyperglycaemia-induced renal damage. They showed that Nrf2 negatively regulates TGF- β 1 and its downstream effectors, such as ECM production (Jiang et al., 2010). Enhanced expression of Nrf2 induced by esculletin might be one of mechanisms of HP renal protection. It has been reported that Nrf2 can induce IL6 expression via an antioxidant response element within the IL6 promoter (Wruck et al., 2011). Within 24 h of incubation under HG condition, oxidative stress is the most harmful effect in mesangial cells, and higher IL6 might exhibit protective effect against anti-oxidation (Mitazaki et al., 2013). In the present study, as shown in Fig. 7, esculletin significantly increased IL6 expression. All these results suggest that esculletin plays an important role in anti-oxidation. Umbelliferone also exhibits anti-oxidation effect to some extent. It also increases the mRNA level of Nrf2, but its effect is weaker than that of esculletin. Several studies have already suggested that umbelliferone and esculletin have anti-oxidation effect in several *in vitro* and *in vivo* oxidation-stress-related models (Sharma et al., 2008; Zhu et al., 2016).

The TGF β -smad signaling pathway plays a key role in end-stage fibrosis both in glomerulosclerosis and interstitial fibrosis (Wang et al., 2015). In the present study, by PAS staining, the less serious interstitial fibrosis was observed in the renal tubular area in HP-treated diabetic rats, and immunohistochemistry also demonstrated that HP can inhibit the deposition of fibronectin and collagen IV in glomeruli and renal interstitium. Western blot suggested that HP treatment significantly reduced the TGF β 1 level in the kidney tissues, which is one of the possible mechanisms by which HP ameliorates renal fibrosis. Under TGF β 1 stimulation, tubular epithelial cells can change fibroblast cells by epithelial-mesenchymal transition (EMT), which is more conducive for the production of ECM. By using HK2 cells, human tubular epithelium cells, under TGF β 1 stimulation, HP, skimmin, and its two metabolites showed the ability to lower fibronectin secretion. The evaluation of its down-stream signaling pathway showed that the phosphorylation of smad2 was inhibited by HP and the three compounds, especially esculletin. Furthermore, the phosphorylation of smad3 was also inhibited by umbelliferone and esculletin. The phosphorylation of smad2/3 is critical in TGF- β -induced EMT (Li et al., 2011); TGF β -induced heteromeric complexes of smad 2, 3, and 4; and their concomitant translocation to the nucleus, which is required for efficient TGF β signal transduction (Nakao et al., 1997). By inhibiting smad2/3 phosphorylation, HP and its metabolites exhibited strong inhibitory effect on fibronectin secretion, and this might be one of possible molecular mechanisms involved in the activity of HP. Owing to their chemical structure with one or two active phenolic hydroxyl groups, umbelliferone and esculletin have strong inhibitory effect.

In conclusion, HP can significantly slow down the progression of DN by improving the renal function and pathophysiological alterations. Its pharmacological efficiency is mainly due to its anti-oxidation and anti-fibrosis activities. Pharmacokinetics demonstrated that the major metabolites of HP, umbelliferone and esculletin, combined with skimmin form the possible material basis of HP activity. Combination of these compounds exhibits renal protective effect possibly by enhancing the expression of Nrf2, inhibiting the phosphorylation of smad2/3, and inhibiting the secretion of extracellular matrix. Further studies are necessary to clarify more detailed mechanisms.

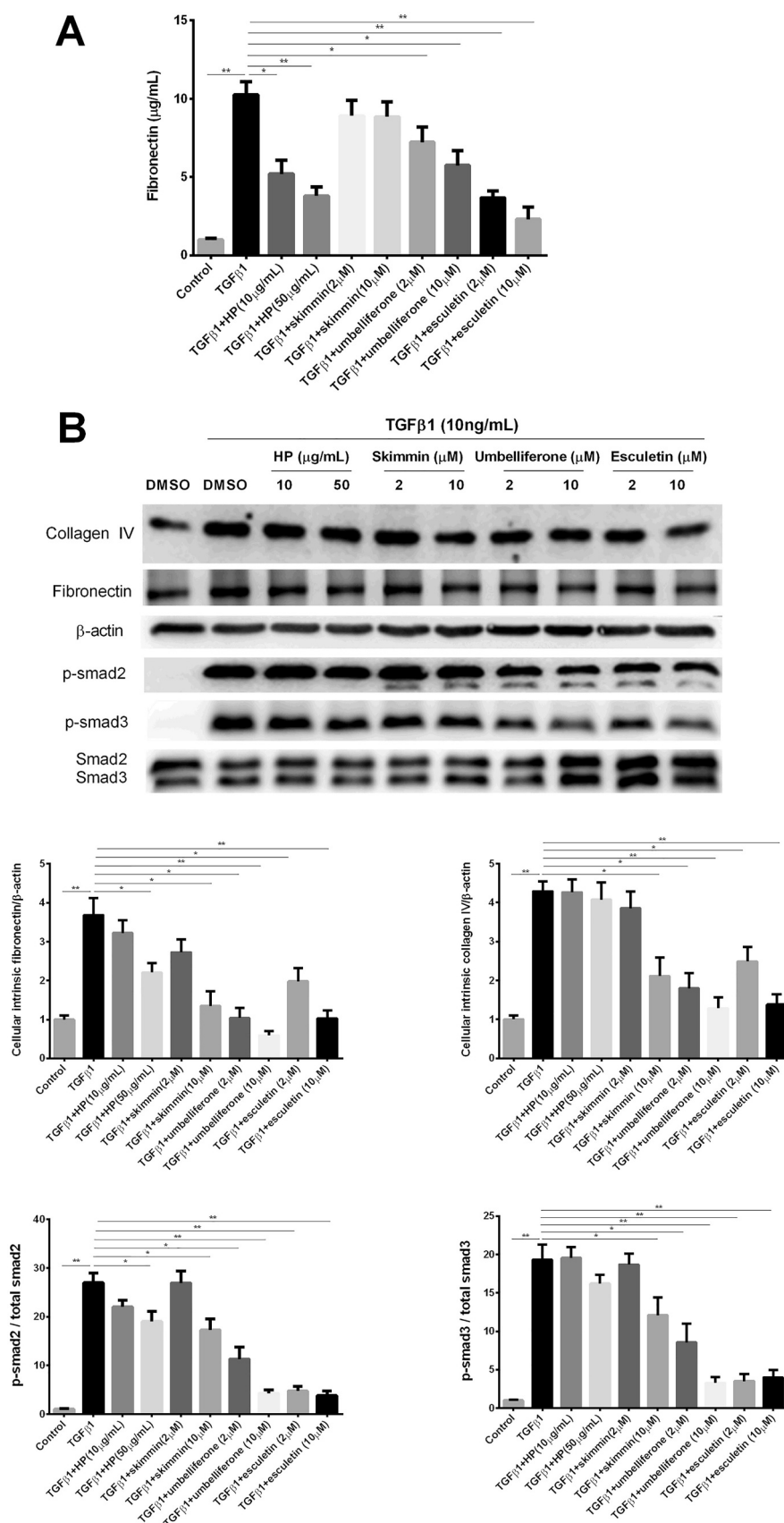


Fig. 8. HP, skimmin, umbelliferone, and esculetin significantly inhibited the secretion of fibronectin in HK2 cells under the stimulation of TGFβ1 by reducing the phosphorylation of smad2/3. (A) Fibronectin concentration in HK2 cell culture medium with TGFβ1 (10 ng/ml) examined by the ELISA. (B) Western blot and quantitation analysis of intrinsic fibronectin and collagen IV normalised by β-actin; the ratio of phosphorylated smad2/3 expression level divided by total smad2/3 expression level. The data are expressed as mean ± SEM of three independent experiments. * $p < 0.05$, ** $p < 0.01$.

Conflict of interest

We confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

Authors' contributions

ZS was in charge of most experiments and manuscript preparation; MJ and ZD were responsible for HP preparation; SL and WD were responsible for pharmacodynamic study; ChX was responsible for the experiment design and funding support.

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